

Solution Requirements

Date	26 June 2026
Team ID	6a3a555ae5895b2c9759029d
Project Name	EduGenie - Google Gemini Powered Learning Assistant
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Single user profile initialized locally via settings; no user auth required for demo.	Low
2	Authorization levels	All users have equal access to ask, explain, quiz, summarize, and recommend modules.	Low
3	External interfaces	Connects to Google Gemini API via HTTPS client; falls back to local HuggingFace T5 model.	High

		<i>services the system must connect to]</i>	
4	Transactions processing	Queries and AI responses are processed as async requests and logged to JSON files.	High
5	Reporting	Interactive browser dashboards displaying query history, quiz scores, and learning roadmaps.	Medium
6	Business rules, etc.	Gemini responses must match expected JSON schemas; fallback triggers automatically if parsing fails.	High
7	Compliance to laws or regulations	No personal data stored; logs are stored locally in data/*.json files, complying with basic privacy.	Low
8	<i>Other</i>	Light/dark theme toggle state persists in localStorage across browser sessions.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria

1	Performance & Speed	FastAPI routing response overhead < 10 ms; Gemini API latency depends on network.	Page load time < 2s
2	Scalability	FastAPI handles high concurrent request volume using async event loop.	Up to 100 concurrent requests
3	Security & Data Privacy	API keys loaded securely via environment variables; input fields sanitized.	Zero hardcoded secrets
4	Reliability & Availability	Dynamic fallback to local offline T5 model / mock mode if Gemini API key is missing.	100% request resilience
5	Usability & Accessibility	Responsive, glassmorphic layout optimized for mobile and desktop screens.	Intuitive single-click navigation
6	Other	Modular Python codebase following clear clean architecture guidelines.	Code readability score > 90%